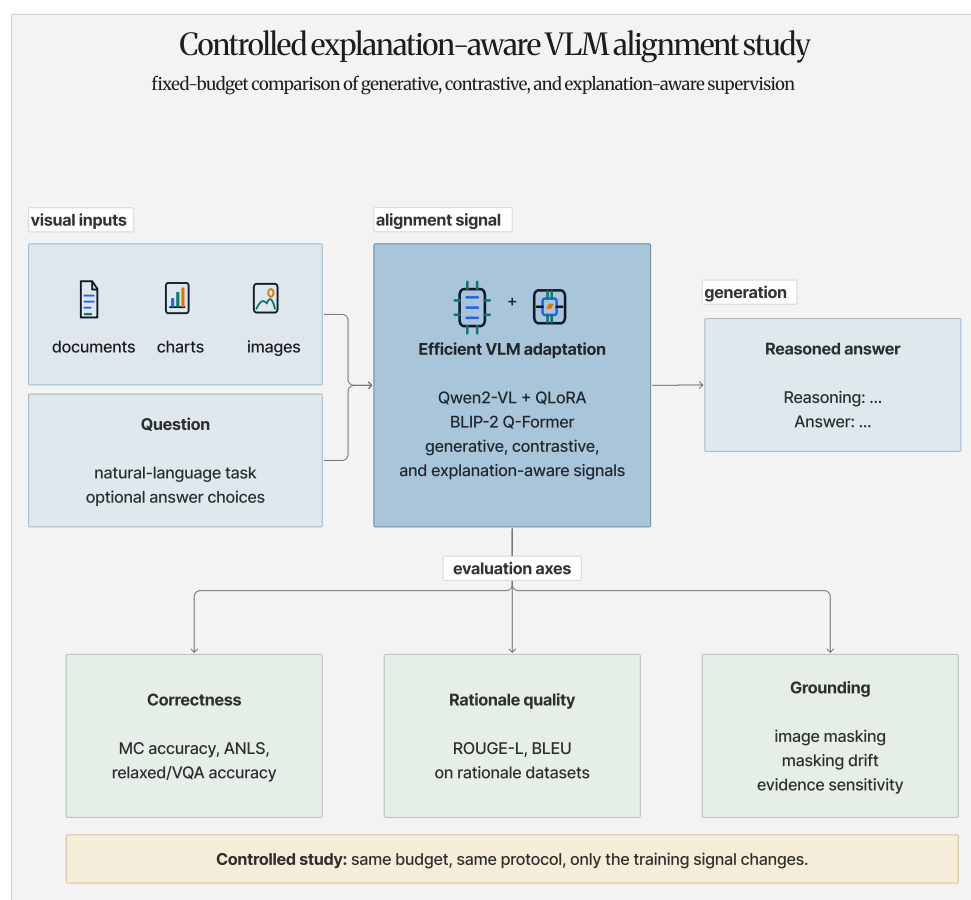


Graphical Abstract

Cross-Modal Generative Alignment for Document, Chart, and Image Reasoning with Explanation-Aware Vision–Language Models

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Highlights

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- The study runs zero-shot, answer-only, rationale-generative, explanation-aware, and BLIP-2 contrastive-enhanced controls under the same single-GPU environment.
- In rationale-bearing datasets, the explanation-aware training does help, but it’s not the best; answer-only controls are strong.
- BLIP-2 contrastive-enhanced fine-tuning yields only a small gain in ScienceQA with BLIP-2 over generative fine-tuning, and the retrieval diagnostic are very close to frozen and random controls.
- Evidence-masking drift shows evidence sensitivity of the model, so the answer gains, rationale quality, and grounding claims are kept separate.

Cross-Modal Generative Alignment for Document, Chart, and Image Reasoning with Explanation-Aware Vision–Language Models

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Abstract

A vision-language model (VLM) is able to read a chart, document, scanned form, or natural image and answer questions about it. But how do we know if the answer is actually derived from the image itself, rather than being a lucky guess on a question based on the model’s remembered language patterns? The correctness of the answer itself is not enough if the reasoning for that answer is not grounded in the visual evidence of the image. Charts and documents are a good example of this problem, because to answer the question correctly, the model has to actually ground its reasoning on specific parts of the image, making it easier to spot when the model is looking at the wrong thing. Although recent work has made progress on this issues, there was no isolation of the training objective as the main variable, with all other factors such as the model, the data, and the budget held constant. Contrastive, generative, and explanation-based methods have all been proposed, but they have not been directly compared under the same conditions, so it is not clear what the objective itself contributed to the final performance. In this paper, we keep the pipeline fixed and change only the training signal, so we can see what the objective does to answer accuracy, explanation quality, and evidence sensitivity when the model, the data, and the compute budget are all held steady. We compare five settings head to head rather than treating them as unrelated systems: BLIP-2 generative fine-tuning, BLIP-2 contrastive-enhanced fine-tuning, Qwen2-VL under answer-only, rationale-generative, and explanation-aware supervision. All the runs use the same resource-constrained single-seed protocol, and the same eval-

uation code as much as possible, across ScienceQA, A-OKVQA, ChartQA, DocVQA, and a VQAv2 subset. The picture that emerges is mixed. On ScienceQA, Qwen2-VL-2B reaches 0.800 accuracy with answer-only supervision, 0.780 in both rationale-generative and length-aware explanation control settings, and 0.765 with fixed-alpha explanation-aware training. On A-OKVQA, the same model reaches 0.830 with answer-only supervision, 0.790 in both rationale-generative and length-aware explanation control training, and 0.795 with fixed-alpha explanation-aware training. ChartQA, DocVQA, and the VQAv2 subset have no gold rationales, so those runs fall back to answer-only controls; they lift the native headline metrics, but they do not test rationale supervision at all. BLIP-2 barely moves: the contrastive objective gives a small gain compared to the generative one on ScienceQA 0.480 vs 0.475, but the retrieval diagnostic stays near frozen and random controls in both cases. Scale is the exception. With a fixed explanation-aware objective on ScienceQA, Qwen2-VL-7B reaches 0.885 while the 2B version reaches 0.765. The masking analysis tells us whether the model relies on the image, not whether its explanation is faithful. Overall, the results do not show a clear win for explanation-aware training. The rationale supervision can help on a single-GPU budget, but the gain is easy to mis-credit to scale or tuning. Hence, it only stands up when answer-only controls, objective balance, model scale, transfer and evidence sensitivity are each reported separately.

Keywords: vision-language models, cross-modal alignment, explanation-aware training, chart and document understanding, faithfulness, parameter-efficient fine-tuning

1. Introduction

Vision-language models (VLMs) are becoming a default tool for answering questions based on the content of images, documents, and charts. The process looks simple: you give it an image and a question, and it will respond with an answer the image actually supports. What is considered as evidence depends entirely on the picture. In a document it might be sitting in a line of a body text, in a table, or in a stamped region of the scanned page. In a chart it might be in an axis label, the relative height of the bars, a colour, or a trend that is not written down explicitly in words. In science and commonsense visual question answering (VQA), it is often not in the image at all, forcing the model to deduce the answer from the background knowledge

it has already learned. Across every one of these, a correct-looking answer is not enough if the reasoning for that answer is not actually based on what is visible in the image. The VLMs are very good at generating the right-looking answers while looking at the wrong thing. That is why in this paper we do not rely on the answer alone, but also check the quality of the explanation and the sensitivity of the answer to visual evidence.

The issue behind all of this is a cross-modal alignment problem: how do we connect the visual signal to the language model in a way that actually makes the two talk to each other? It turns out that the quality of that wiring depends on many factors, such as the visual resolution, the data mixture, the connector design, the instruction tuning, and the adaptation method [1, 2, 3]. Open model families are very helpful, because backbones such as Qwen2-VL, BLIP-2, and LLaVA-OneVision are strong enough out of the box, while still being adaptable on an academic budget [4, 5, 6]. At the same time, document and chart studies show that general visual instruction following is not enough when it comes to text-heavy and structured images [7, 8, 9, 10].

Furthermore, there is also a budget reality to consider. Most students and small labs cannot afford full fine-tuning of a large multimodal model. Methods like DoRA (weight-decomposed low-rank adaptation) make it possible to fine-tune the model by updating only a small portion of the parameters, while keeping the base model frozen or quantised [11]. Hence, in this study, we take the same approach using low-rank adaptation (LoRA) and its quantised variant (QLoRA) instead of full-model fine-tuning [12, 13]. The main goal is not scoring the highest on a leaderboard, but rather to check the model’s performance when only the alignment objective changes while keeping all other factors constant.

1.1. Gap Analysis

The gap is isolation. Most papers change the model, the data, the training size, and the objective all at once. Once the final result is reported, it is hard to tell how much the objective itself contributed. Maybe the objective helped, but it might also be the case that it was simply a bigger model, cleaner data, or more tuning and not necessarily the objective itself.

Moreover, each objective has its own blind spot. Contrastive learning is good at matching an image to a caption, but it does not teach the model to generate an explanation or an answer from the image. Generative learning is good at producing an answer, but it does not necessarily mean that the

48 model is basing its answer on the correct parts of the image. Explanation-
49 aware training teaches the model to actually write reasoning to come to the
50 answer, but the model could have learned to write a convincing explanation
51 after the fact, which might not actually be the cause of the answer.

52 Hallucination and faithfulness evaluation has mostly been studied on nat-
53 ural images, where the common test is whether the model names objects that
54 are not there. Charts and documents have recently come into focus too, with
55 benchmarks for chart hallucination, chart grounding, and document-answer
56 groundedness emerging [14, 15, 16]. They are still comparatively underex-
57 plored, even though their evidence is more structured and easier to verify.
58 That is exactly what makes them a good test case for grounding.

59 These gaps are not closed by a new model or benchmark. What is actually
60 missing is a controlled study that holds the model, the data, the budget,
61 and the evaluation procedure constant while changing only the alignment
62 objective, to test not only the answer quality but also the explanation quality
63 and the evidence sensitivity of the model under each objective.

64 1.2. Research Questions

65 The main question is to investigate how the resource-constrained vision-
66 language model should be aligned to achieve both accurate and faithful visual
67 reasoning. The following research questions break that down into more mea-
68 surable parts.

- 69 1. RQ1: How have vision-language alignment methods evolved for
70 grounded visual reasoning and explanation generation?
- 71 2. RQ2: How does BLIP-2 generative fine-tuning compare with BLIP-2
72 contrastive-enhanced fine-tuning on visual reasoning?
- 73 3. RQ3: Does explanation-aware fine-tuning improve answer accuracy and
74 explanation quality on datasets with gold rationales?
- 75 4. RQ4: How does the same fine-tuning framework perform across docu-
76 ment, chart, science, commonsense, and natural-image domains?
- 77 5. RQ5: How sensitive are the model’s answers to visual evidence under
78 image-masking perturbations?
- 79 6. RQ6: How does the selected explanation-aware QLoRA setup scale
80 from Qwen2-VL-2B to Qwen2-VL-7B?
- 81 7. RQ7: How well do fine-tuned adapters transfer across visual domains
82 and prompt formats?

83 1.3. Problem Statement

84 The problem is cross-modal alignment in a resource-constrained environ-
85 ment. Given the same pretrained vision–language model, we analyse the
86 effect of different training objectives on answer accuracy, explanation qual-
87 ity, and evidence sensitivity. The model, the data format, the optimiser, and
88 the evaluation process are kept constant wherever possible, leaving only the
89 training signal as the variable. The BLIP-2 pair reuses one Q-Former train-
90 ing setup. The Qwen2-VL generative and explanation-aware runs reuse one
91 QLoRA setup. The goal is not to find the best scoring setup, but to observe
92 whether the chosen objective leads the model in a direction that matters for
93 a more trustworthy reasoning, and to check whether the answer quality, the
94 explanation quality, and the evidence sensitivity move together or separately
95 under each objective.

96 1.4. Novelty of this study

97 The contribution is the controlled comparison, not a new architecture.
98 Instead of introducing a new architecture, we compare existing ones under
99 different training objectives. We consider the faithfulness of the model’s
100 reasoning as a first-class result, rather than concentrating solely on answer
101 accuracy. The paper contributes the following:

- 102 • A comparison of zero-shot, answer-only, BLIP-2 generative, BLIP-2
103 contrastive-enhanced, Qwen2-VL rationale-generative, and Qwen2-VL
104 explanation-aware alignment under the same data and optimisation
105 settings.
- 106 • Evaluation along three separate axes, namely answer quality, explana-
107 tion quality, and evidence sensitivity, across document, chart, science,
108 and commonsense tasks.
- 109 • For multiple-choice reasoning, a generate-then-score protocol is used,
110 so the model is evaluated only after it has produced its own reasoning
111 trace.
- 112 • A direct test of how QLoRA behaves with scale, instead of assuming
113 that full fine-tuning or very large models are the only things worth
114 studying.

115 1.5. *Why this study matters*

116 The usefulness of this study is in the constraints. It runs on a resource-
117 constrained setup that is actually accessible to students and small labs. It
118 also avoids concentrating solely on a single aspect of performance, and in-
119 stead keeps answer accuracy, explanation quality, and evidence sensitivity as
120 separate readings. On documents and charts especially, that separation is
121 particularly important, because the evidence should trace back to a visible
122 region. The whole pipeline is reusable: every dataset is transformed into one
123 shared format, and every experiment is declared in a configuration file, so the
124 code can be used for future comparisons of new objectives, new models, or
125 new datasets, without worrying about hidden differences between runs. And
126 the results are honest rather than overselling: the rationale supervision does
127 help on the science and commonsense runs, but the answer-only controls stay
128 strong, and the BLIP-2 contrastive objective provides only a small answer
129 gain on top of weak retrieval evidence.

130 2. Literature Review

131 The review covers relatively latest works in the areas of vision–language
132 model design, text-rich and document-heavy images, charts and structured
133 visual reasoning, faithfulness and critique, and efficient adaptation. We group
134 it by the parts that actually matter here the most: model design, text-
135 rich images, chart reasoning, faithfulness, datasets, metrics, and low-budget
136 adaptation.

137 2.1. *Recent vision–language model design*

138 Recent vision–language models are not built around a single alignment
139 trick. Their performance is a result of many choices working together, such as
140 the backbone architecture, the visual tokenisation method, the data mixture,
141 the connector design, and the instruction tuning. Since all those factors are
142 usually changed together, it is hard to tell which part actually caused an
143 improvement in the final performance.

144 This is exactly the issues raised by recent ablation studies, which demon-
145 strate that these factors are still poorly separated, and that it is vital to
146 conduct controlled comparisons [1]. MM1.5 also makes the same argument,
147 showing that optical character recognition (OCR) data, synthetic captions,
148 instruction mixtures, and specialised variants can each weigh as much as raw
149 scale in the final performance of a vision–language model [2]. Molmo and

150 PixMo are useful here because they provide open weights and open data that
151 let us actually read the recipe and understand what is going on under the
152 hood [3]. LLaVA-OneVision shows that a single family of models can be used
153 across image and video when the data mixture is chosen with care [6].

154 For this study, the conclusion is simple. If the model, the data, and the
155 budget are all changing together, then the contribution of the objective itself
156 is hard to tell. That is why we need a controlled comparison, where the
157 model, the data, and the budget are held constant while only the training
158 signal changes.

159 Open model families are also important, because they enable us to ac-
160 tually run the controlled comparison on a practical budget. Qwen2-VL is
161 designed to work with dynamic image resolution and text-heavy visual in-
162 puts such as documents, charts, and diagrams [4]. BLIP-2’s Q-Former lets us
163 test an image–answer contrastive loss while keeping the larger vision encoder
164 and language model frozen [5]. We use these open models because they make
165 controlled, parameter-efficient experiments feasible under limited resources.

166 2.2. *Text-rich and document-heavy images*

167 Documents are difficult for a specific reason. The answer in this case is
168 not in the whole image, but rather in a small part of it, such as a line of text, a
169 table cell, or a layout region. Hence, the task is not just an image recognition
170 problem. The model has to read the document and find the evidence in the
171 right place, which is a different skill than naming objects in a natural image.

172 Recent work shows that text-rich image reasoning is moving toward mul-
173 timodal large language models, especially for OCR-heavy and layout-heavy
174 inputs [7]. DocLLM is a good example of this trend. It encodes the document
175 layout directly into the generative document-understanding problem, which
176 is a step toward more faithful reasoning on documents [8]. OmniDocBench
177 focuses on the coverage angle instead, scoring diverse PDF parsing cases
178 with detailed annotations, which is a step toward more structured evalua-
179 tion of document understanding [17]. In this study, we use DocVQA as a
180 test case for document understanding, because it is a well-known benchmark
181 where the answer is usually located on scanned text, layout, and local visual
182 context [18].

183 Our goal is not to build a new document model. We want to see how
184 much the training objective itself contributes to the model’s ability to read
185 documents and answer questions about them.

2.3. Charts and structured visual reasoning

Charts create a different kind of problem. In this case, the model has to read the labels, understand the axes and tick marks, and determine the relationships between the visual elements, such as bars, lines, or points. ChartInstruct and ChartGemma demonstrate that instruction tuning on chart images can help in both chart understanding and reasoning [9, 10]. ChartMimic shows the same point by training a model to generate code for that chart, which makes the model understand the chart in a more structured way [19]. ChartMuseum shows that current large visual-language models still perform far poorer than humans on chart reasoning [20]. MathVista, MMMU-Pro, and MMReason extend this concern to mathematical and multi-step reasoning, which is often required for chart understanding [21, 22, 23].

This is why we use ChartQA, ScienceQA, and A-OKVQA in the same study [24, 25, 26]. Together, they cover structured chart reasoning, science reasoning, and knowledge-heavy visual reasoning.

2.4. Faithfulness, critique, and efficient adaptation

Explanation-aware training has a subtle problem. The model might learn to write an explanation that looks convincing, but that is not actually based on the image and is not the real cause leading to the answer. Recent studies approach this problem from different angles. MMMU tests expert-level multimodal reasoning across many visual formats, to see if VLMs can do expert-style visual reasoning and not just simple image captioning [27]. MMStar checks whether a model actually needs the image to answer the question or whether the answer can be derived from the learned knowledge alone [28]. HallusionBench focuses on cases where language hallucination and visual illusion appear together in LVLM responses [29]. Hallusination-augmented contrastive learning turns a hallucination into a contrastive signal, to teach the model to avoid it [30]. VISCO and Critic-V evaluate the model’s ability to critique, correct, and detect errors in its own output [31, 32]. These works suggest that faithfulness should be measured separately from answer accuracy.

Efficient adaptation is also important, because it allows us to test the effect of the training objective without needing to fine-tune the entire model. DoRA is a good example of this approach, which demonstrates that we can actually adapt the model by updating only a small portion of the parameters [11]. Reason-RFT points to another direction that is relevant here, where post-training specifically rewards visual reasoning [33]. In this paper,

we stay with supervised objective comparisons, but the concern is similar. We evaluate the model as a reasoning system, rather than only as an answer generator.

Table 1 summarises the relevant works and the gaps we address in this study. It is not meant to be a comprehensive review of the field, but rather a snapshot of the recent trends and how they relate to our controlled comparison.

Table 1: Representative related work and how it connects to the gap studied here. A check mark means the line of work directly covers the property, a cross means it is not part of the main scope, and a tilde means partial coverage.

Work	Main focus	Generation	Structured VQA	Rationales	Efficient FT	Connection to this study
Recent VLM construction [1, 3, 6, 2]	Model and data recipe	✓	~	~	~	Shows that model design and training data must be controlled before we can attribute gains to the objective.
Open VLM backbones [4, 5]	Adaptable model families	✓	~	~	~	Provide practical backbones for controlled experiments under a fixed compute budget.
Text-rich document work [7, 8, 17]	Documents and PDF parsing	✓	✓	~	~	Motivates the document setting, but usually does not compare training objectives under one fixed backbone.
Chart reasoning work [9, 10, 20, 19]	Chart understanding	✓	✓	~	~	Shows that chart reasoning is still difficult and needs structured evaluation beyond ordinary image QA.
Recent reasoning benchmarks [21, 27, 28, 22, 23]	Harder multi-modal evaluation	~	✓	~	✗	Motivate evaluation on visual, mathematical, expert-level, and multi-step reasoning tasks.
Faithfulness and critique [29, 30, 31, 32]	Reliability of reasoning	~	✓	✓	✗	Support the need to measure hallucination, critique, and evidence use separately from answer accuracy.
DoRA / Reason-RFT [11, 33]	Efficient and reasoning-focused adaptation	~	✗	~	✓	Motivate low-resource adaptation and explicit reasoning-oriented post-training.
This study	Controlled explanation-aware alignment	✓	✓	✓	✓	Compares objective choices directly and reports answer accuracy, rationale quality, and evidence sensitivity together.

A common pattern emerges from this review. Earlier work on multimodal alignment focused mainly on learning image–text representations, or on generating an answer from the image and the question. More recent work has started to pay attention to the faithfulness of the model’s reasoning, and to the quality of the explanations it generates. However, it is still unclear how much the training objective itself contributes to the model’s performance. In this study, our goal is to fill that gap by conducting a controlled comparison of different training objectives while keeping all other factors constant.

3. Methodology

One rule drives the method: hold the pipeline fixed, change the objective. A dataset goes in and is converted to a shared format. From there the model

is loaded, QLoRA adapters are attached if the run needs them, the chosen objective is trained, and evaluation goes through the same generate-then-score procedure every time. Figure 1 lays out the full pipeline. The design is conservative on purpose. For what we are asking, one clean controlled comparison beats a sprawling leaderboard-style run. Every experiment is declared in a configuration file: the dataset, the backbone, the objective, the adapter settings, the metrics, and the seed, all on the record. One implementation path handles baseline evaluation, fine-tuning, and final evaluation. That is what keeps hidden differences from creeping in between runs.

3.1. Dataset

Five public benchmarks carry the experiments. DocVQA stands in for documents, where the answer usually rides on scanned text, layout, and local visual context [18]. ChartQA covers chart reasoning, where it can turn on a label, an axis, a value, or a comparison [24]. ScienceQA brings multimodal science questions with answer choices and written explanations, which is what makes it usable for rationale supervision [25]. A-OKVQA adds commonsense questions and rationales over natural images [26]. VQAv2 plays the general natural-image control [34]. One caveat there: the completed run draws on the configured `Erland/VQAv2-sample` source, capped at 900 training and 100 evaluation examples, so we report it as a subset result, not a full VQAv2 score. Whatever the source, every dataset is bent into one schema: image, question, optional answer choices, target answer, optional rationale. A dataset with no gold rationale is treated as answer-only, unless a separate rationale experiment is explicitly marked.

Table 2: Benchmarks used in the study. The sizes are rounded public benchmark sizes before filtering by image availability, answer type, or rationale availability. The completed experiments use capped train and evaluation splits so that every run fits the same practical compute budget.

Dataset	Domain	Approx. size	Answer type	Gold rationale	Primary metric
DocVQA [18]	Document	50K	short text span	no	ANLS
ChartQA [24]	Chart	32K	numeric / short text	no	relaxed accuracy
ScienceQA [25]	Science image	21K	multiple choice	yes	accuracy
A-OKVQA [26]	Commonsense image	25K	multiple choice / short text	yes	accuracy
VQAv2 subset [34]	Natural image	900/100 run split	open ended	no	VQA accuracy

3.2. Detailed methodology

Figure 2 shows the model-level setup. Every backbone sits behind one common interface, so the training code never forks into a separate path per

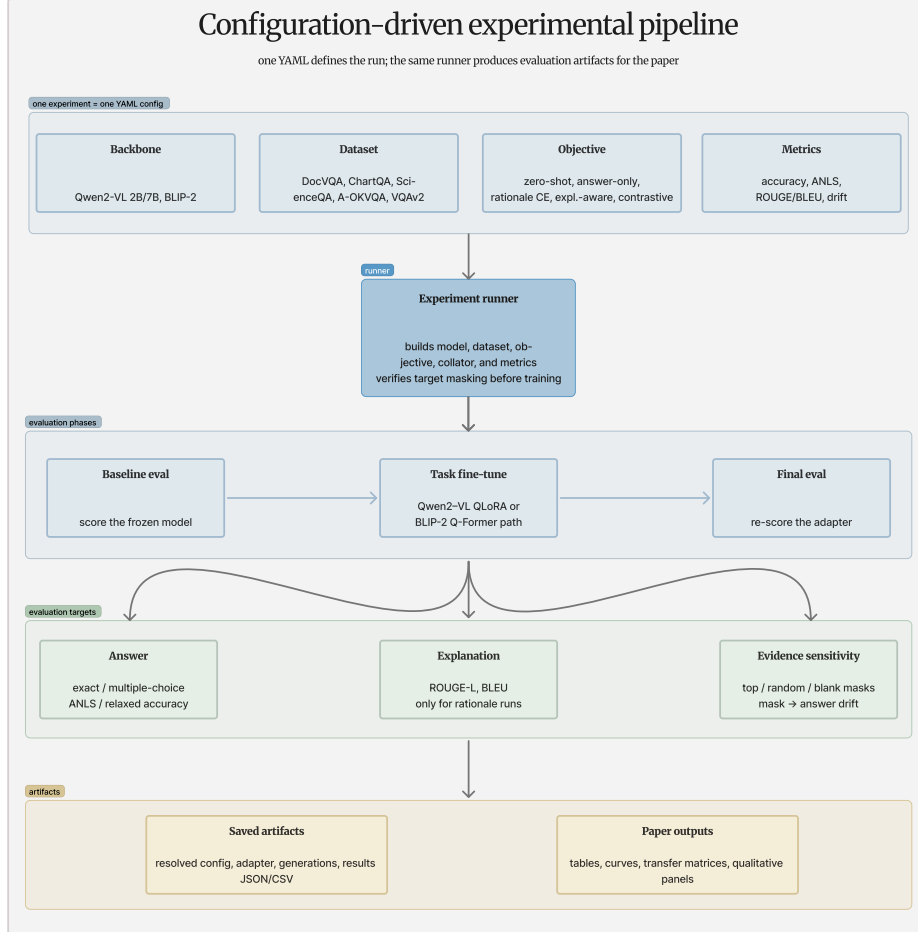


Figure 1: Overall workflow of the proposed explanation-aware alignment study. Datasets are normalised into a shared multimodal example format. Qwen2-VL runs are adapted with QLoRA, BLIP-2 runs train the Q-Former-side alignment path, and each objective is evaluated through answer, rationale, and evidence-sensitivity metrics.

268 model. Qwen2-VL loads in 4-bit precision and picks up rank-16 QLoRA
 269 adapters [12, 13]. On the BLIP-2 contrastive arm, the vision encoder and
 270 the OPT (Open Pre-trained Transformer) language model are frozen solid.
 271 Only the Q-Former-side components, the language projection, and a small
 272 answer-projection head ever update. The baseline variant just evaluates the

Table 3: Input construction examples. The complete token, label, span, and collator traces are saved as pipeline artifacts; this table gives the paper-level view of what each prompt family means.

Prompt family	Raw fields	Prompt seen by model	Supervised target	Supervised spans
Answer-only	image, question, answer	Question: ... plus choices when available	Answer: <answer>.	answer tokens only
Rationale-generative	image, question, answer, gold rationale	Question: ... plus choices when available	Reasoning: <rationale> Answer: <answer>.	full target under uniform CE
Explanation-aware	image, question, answer, gold rationale	Question: ... plus choices when available	Reasoning: <rationale> Answer: <answer>.	rationale and answer spans weighted separately
Answer-only fall-back	image, question, answer, no gold rationale	Question: ...	Answer: <answer>.	answer tokens only

model before any task fine-tuning. The contrastive-enhanced variant bolts on an auxiliary image-answer term: it tugs the pooled Q-Former image representation toward the correct answer and away from the others in the batch. The generative variant trains the model to produce the answer from the image and question. The explanation-aware variant trains it to produce a rationale first, then the answer. So the variants differ in one thing only, the supervision signal, and everything else inside a comparison is held as close as it can be.

In generative training, prompt and target live in one concatenated sequence, but the loss only ever touches the target tokens. That detail is easy to get wrong, because multimodal processors slip image placeholder tokens in front of the text. Count the prompt length by hand and the supervised span quietly slides off position. The implementation sidesteps it by measuring prompt length with the very same processor that encodes the full sequence. The contrastive objective reuses those span tags, pooling answer tokens alone into the InfoNCE (information noise-contrastive estimation) text representation [35]. Rationale tokens still receive the generative loss; they simply are not treated as the contrastive positive. For the explanation-aware objective, the target splits in two, an explanation span and an answer span. Equation 1 weights the two spans with α . When $\alpha = 1$, only the answer span is weighted, although the target sequence can still contain the gold rationale before the answer. A clean answer-only control is therefore reported separately, using a target template that contains only **Answer:**. When $\alpha = 0$, only the explanation span is supervised. Intermediate values test whether answer and rationale supervision help each other or compete.

$$\mathcal{L} = \alpha \mathcal{L}_{\text{answer}} + (1 - \alpha) \mathcal{L}_{\text{explanation}} \quad (1)$$

where $\mathcal{L}_{\text{answer}}$ and $\mathcal{L}_{\text{explanation}}$ are the masked cross-entropy losses over the

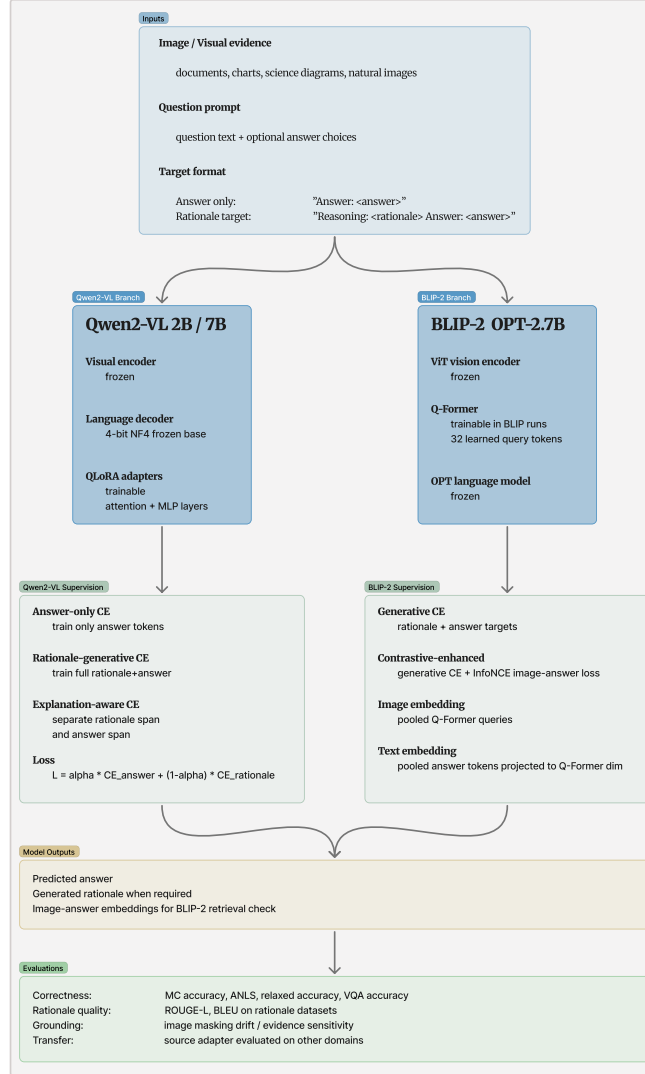


Figure 2: Model-level architecture for explanation-aware cross-modal alignment. Visual evidence and the question prompt are encoded into a multimodal token stream. The Qwen2-VL path uses trainable QLoRA adapters and separate rationale and answer supervision spans. The BLIP-2 path keeps the vision tower and OPT language model frozen and trains the Q-Former-side generative or contrastive alignment components.

299 answer and explanation token spans, and α controls their relative weight.
 300 There are two alpha modes. Fixed-alpha mode treats α as a dial we set
 301 by configuration, an ablation knob. Length-aware mode does something

different: it derives the effective answer weight straight from the supervised token counts:

$$\alpha_{\text{eff}} = \frac{\lambda n_{\text{answer}}}{\lambda n_{\text{answer}} + n_{\text{explanation}}}, \quad (2)$$

where n_{answer} and $n_{\text{explanation}}$ are the answer and explanation token counts after collation, and λ is the answer-weight multiplier. With $\lambda = 1$, this collapses to plain uniform token-level cross-entropy over the rationale-plus-answer target. So the fixed-alpha sweep is the ablation; the length-aware run is the audit control for the natural, token-weighted setting.

Multiple-choice datasets are scored in two steps at inference. First the model writes a short reasoning trace from the image and question. Then, in the context of that trace, each candidate answer gets a length-normalised conditional likelihood, and the highest score wins. The two-step design is there to avoid brittle string parsing and option-label copying. Open-ended datasets are simpler: the generated answer string is scored with the dataset’s native metric. Whenever the model produces a rationale, we save it, and those saved rationales feed the text-quality and faithfulness analysis.

3.3. Evaluation metrics

Evaluation splits into three parts: answer quality, rationale quality, and evidence sensitivity. Answer quality always uses the benchmark’s own native metric. When the tables say “headline metric,” they mean exactly that native primary metric for the target dataset, not some universal number to be read straight across datasets. For ScienceQA and A-OKVQA that metric is multiple-choice accuracy. DocVQA uses average normalised Levenshtein similarity (ANLS), per the benchmark convention [18]. ChartQA uses relaxed accuracy on numerical answers [24]. The VQAv2 subset uses VQA consensus accuracy wherever annotator answers exist [36, 34]. Rationale quality leans on BLEU and ROUGE-L whenever gold rationales are available [37, 38]. Neither is a perfect judge, but together they give a basic lexical and sequence-level check on the explanation. Evidence sensitivity asks a blunter question: does the answer’s likelihood move when we mask part of the image? This borrows the old idea of occlusion sensitivity analysis [39]. The specific measure is evidence-masking consistency, written out in Equation 3. The qualitative diagnostic then lines up the high-drift regions against whatever visual evidence the generated explanation actually names, when that comparison is possible.

$$\text{drift}(r) = s(a \mid I) - s(a \mid I \setminus r) \quad (3)$$

where $s(a \mid I)$ is the length-normalised log-likelihood of the gold answer a given image I , and $I \setminus r$ is the same image with region r occluded. A large positive drift says the masked region mattered. A drift near zero says the opposite: the answer probably leaned on language priors, a shortcut, or non-visual context. The absolute scale does not travel across datasets, since answer length, prompt family, image resolution, and task format all bend the likelihood scores. The split it captures is the familiar one between a useful explanation and a faithful one. It also speaks to recent hallucination and critique work, where a model happily describes or relies on visual content that simply is not there [29, 30, 31, 32]. That is why evidence sensitivity sits next to answer accuracy here, rather than being parked as a post-hoc example.

3.4. Experimental settings

Every variant runs under the same fixed-budget protocol, and any deviation from it is written into the experiment ledger. The Qwen2-VL runs share one recipe: AdamW, a learning rate of 2×10^{-4} , cosine decay, three percent warm-up, weight decay of 0.01, bf16 compute, and an effective batch size of 32 reached through gradient accumulation. The base model loads under 4-bit NormalFloat (NF4) quantisation. Unless a result table says otherwise, the QLoRA adapters use rank 16 and scaling factor 32 [12, 13]. The workhorse backbone is Qwen2-VL-2B-Instruct, strong enough for the target tasks yet small enough to fit the budget [4]. For BLIP-2 we use Salesforce BLIP-2 OPT-2.7B, vision tower and OPT language model frozen, training only the Q-Former-side alignment components [5]. The scale ablation reuses the same QLoRA recipe on Qwen2-VL-7B. Each run logs its dataset split, filtered count, seed, objective, adapter configuration, and metrics. The pipeline also writes a split-and-leakage audit from the selected train and evaluation identifiers, wherever the dataset exposes stable IDs. For the capped datasets in the final artifact set, that overlap count comes back at zero. DocVQA is the one non-standard split: the loaded dataset only exposes validation and test, so the controlled run trains on `validation[:80%]` and evaluates on `validation[80%:]`, with native-ID overlap checking. The audit shows zero native-ID overlap, but it cannot prove document-level disjointness past the IDs the loader exposes. So we treat the DocVQA number as a capped internal adaptation result, not a public-test benchmark claim. We also attach

371 approximate uncertainty estimates to the accuracy-style headline metrics.
 372 With most evaluation caps at 200 examples and only a few seeds in the bud-
 373 get, a gap like 0.790 versus 0.795 is an observed difference, nothing more.
 374 Table 4 lists the default configuration.

Table 4: Default Qwen2-VL QLoRA fine-tuning configuration. BLIP-2 contrastive runs use the separately reported frozen-vision/frozen-LLM Q-Former training setup. Any run that changes these values is reported separately in the experiment ledger and in the Results section.

Training Configuration	
Adapter	QLoRA (4-bit NF4, rank 16, scaling 32)
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
Learning rate	2×10^{-4} (cosine, 3% warm-up)
Weight decay	0.01
Effective batch size	32 with gradient accumulation
Precision	bf16 compute
Primary seed policy	one seed per complete configuration

375 3.5. Experiment ledger

376 Before any scores, Table 5 lays out the comparisons. The line that mat-
 377 ters is rationale availability. ScienceQA and A-OKVQA can carry rationale-
 378 generative and explanation-aware supervision, because they ship gold ratio-
 379 nales. ChartQA, DocVQA, and VQAv2 stay on answer-only fallback, plus
 380 the transfer and domain runs, until some future experiment adds rationale
 381 supervision of their own.

Table 5: Main experiment ledger. The full generated ledger includes exact caps, seeds, prompt variants, alpha mode, and configuration hashes.

Comparison	Runs included	Use in the paper
BLIP-2 objective pair	BLIP-2 OPT-2.7B on ScienceQA: generative and generative+contrastive.	Tests whether the auxiliary contrastive objective adds anything beyond BLIP-2 generative fine-tuning.
ScienceQA controls	Qwen2-VL-2B on ScienceQA: answer-only, rationale+answer CE, fixed- α explanation-aware sweep, and length-aware control.	Separates answer supervision, rationale supervision, and alpha weighting on a gold-rationale dataset.
A-OKVQA controls	Qwen2-VL-2B on A-OKVQA: answer-only, rationale+answer CE, fixed- α explanation-aware, and length-aware control.	Repeats the rationale-supervision comparison on a second gold-rationale domain.
Fallback domains	Qwen2-VL-2B on ChartQA, DocVQA, and VQAv2 with answer-only fallback prompts.	Reports in-domain adaptation and transfer for datasets without gold rationales; these rows are not used as explanation-aware evidence.
Scale	Qwen2-VL-2B and Qwen2-VL-7B on ScienceQA with the same fixed explanation-aware QLoRA setting.	Checks whether the selected setup changes when the backbone scale increases.
Transfer	Qwen2-VL-2B source adapters evaluated on target-domain prompts across all datasets.	Tests cross-domain behaviour separately from the official in-domain result table.
Masking	Selected Qwen2-VL runs with image-region perturbation and saved visual examples.	Measures evidence sensitivity beside answer accuracy and rationale quality.

382 4. Results

383 The results track the comparison types from the ledger, in order. The
384 BLIP-2 objective pair comes first, testing whether the auxiliary contrastive
385 term earns its place beyond generative fine-tuning. Then the ScienceQA
386 and A-OKVQA runs pull answer-only, rationale-generative, and explanation-
387 aware supervision apart on the two datasets that have gold rationales. The
388 chart, document, and VQAv2 runs follow, showing in-domain adaptation
389 where no gold rationales exist. After that come masking-based evidence sen-
390 sitivity, the Qwen2-VL scale check, and cross-domain transfer. One reminder
391 throughout: every score comes from the capped single-seed protocol above,
392 so they are controlled framework results, not leaderboard numbers.

393 4.1. Objective-level comparison

394 Table 6 holds the direct objective comparisons. Take the BLIP-2 Sci-
395 enceQA pair first: both trained variants clear the 0.445 frozen baseline, the
396 contrastive-enhanced run landing at 0.480 and the generative control at 0.475.
397 That gap is tiny, and it sits inside the approximate uncertainty intervals in
398 Table 7. Retrieval tells the same cautious story: the contrastive-enhanced
399 model posts recall@1 (R@1) of 0.010, R@5 of 0.020, R@10 of 0.030, and a
400 mean reciprocal rank (MRR) of 0.0307. Qwen2-VL-2B is stronger across
401 the board. Rationale-generative fine-tuning reaches 0.780 on ScienceQA and
402 0.790 on A-OKVQA. On A-OKVQA the explanation-aware run reaches 0.795
403 accuracy, 0.337 ROUGE-L, and 0.087 BLEU, numerically a hair above the
404 generative control, which we read as an observed single-run difference rather
405 than a settled win. The honest surprise is the answer-only controls: 0.800
406 on ScienceQA and 0.830 on A-OKVQA, both strong. The zero-shot col-
407 umn, meanwhile, is one shared backbone-dataset reference for orientation,
408 not a fresh prompt-family baseline per row. Back on ScienceQA, the fixed
409 $\alpha = 0.50$ explanation-aware run reaches 0.765. The length-aware control
410 reaches 0.780, level with the rationale-generative control but still under the
411 answer-only number.

Table 6: Objective-level results under the controlled protocol. The zero-shot reference is the shared pre-fine-tuning score for the same backbone and dataset group, not a separately recomputed frozen score for every prompt family. Accuracy is multiple-choice accuracy for ScienceQA and A-OKVQA.

Backbone	Dataset	Objective	Zero-shot ref.	Final	ROUGE-L	BLEU
BLIP-2 OPT-2.7B	ScienceQA	Generative	0.445	0.475	–	–
BLIP-2 OPT-2.7B	ScienceQA	Contrastive-enhanced	0.445	0.480	–	–
Qwen2-VL-2B	ScienceQA	Answer-only	0.400	0.800	–	–
Qwen2-VL-2B	ScienceQA	Rationale+answer CE	0.400	0.780	0.693	0.559
Qwen2-VL-2B	ScienceQA	Explanation-aware, $\alpha = 0.50$	0.400	0.765	0.643	0.493
Qwen2-VL-2B	ScienceQA	Explanation-aware, length-aware	0.400	0.780	0.696	0.563
Qwen2-VL-2B	A-OKVQA	Answer-only	0.355	0.830	–	–
Qwen2-VL-2B	A-OKVQA	Rationale+answer CE	0.355	0.790	0.329	0.083
Qwen2-VL-2B	A-OKVQA	Explanation-aware, $\alpha = 0.50$	0.355	0.795	0.337	0.087
Qwen2-VL-2B	A-OKVQA	Explanation-aware, length-aware	0.355	0.790	0.330	0.083

Table 7: Approximate uncertainty for selected accuracy-style headline metrics. These intervals use the generated normal-approximation standard errors for capped evaluation sets, so they are a caution about small observed differences rather than a replacement for multi-seed testing.

Run	Metric	Eval cap	Score	Approx. 95% CI
BLIP-2 generative ScienceQA	MC accuracy	200	0.475	0.406–0.544
BLIP-2 contrastive ScienceQA	MC accuracy	200	0.480	0.411–0.549
ScienceQA answer-only	MC accuracy	200	0.800	0.745–0.855
ScienceQA rationale+answer CE	MC accuracy	200	0.780	0.723–0.837
ScienceQA explanation-aware, $\alpha = 0.50$	MC accuracy	200	0.765	0.706–0.824
ScienceQA explanation-aware, length-aware	MC accuracy	200	0.780	0.723–0.837
A-OKVQA answer-only	MC accuracy	200	0.830	0.778–0.882
A-OKVQA rationale+answer CE	MC accuracy	200	0.790	0.734–0.847
A-OKVQA explanation-aware, $\alpha = 0.50$	MC accuracy	200	0.795	0.739–0.851
Qwen2-VL-7B ScienceQA, $\alpha = 0.50$	MC accuracy	200	0.885	0.841–0.929

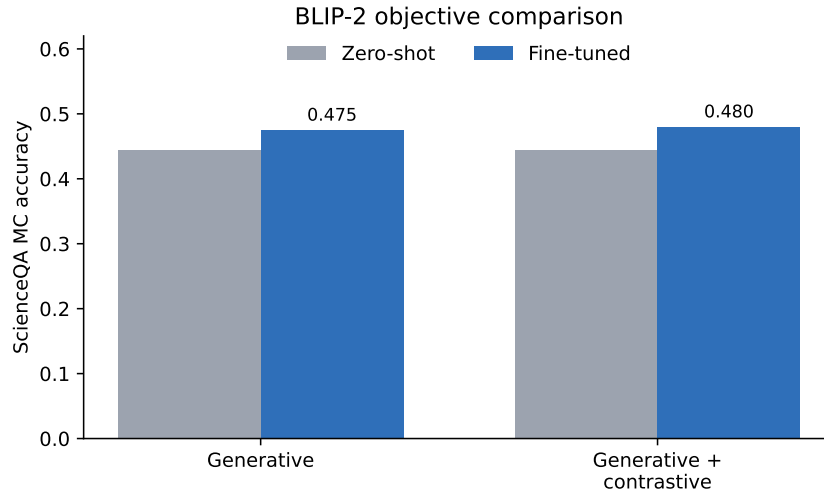


Figure 3: BLIP-2 generative and contrastive-enhanced ScienceQA results. Both runs improve over the frozen baseline, with only a small observed gain from the auxiliary contrastive term.

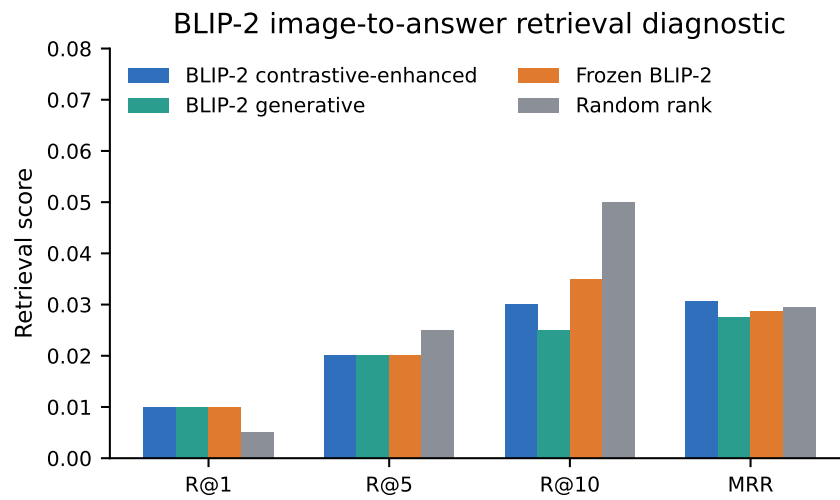


Figure 4: BLIP-2 image-to-answer retrieval diagnostic. The analysis reports frozen BLIP-2, BLIP-2 generative, BLIP-2 contrastive-enhanced, and random-rank controls so the contrastive branch is not interpreted without a reference point.

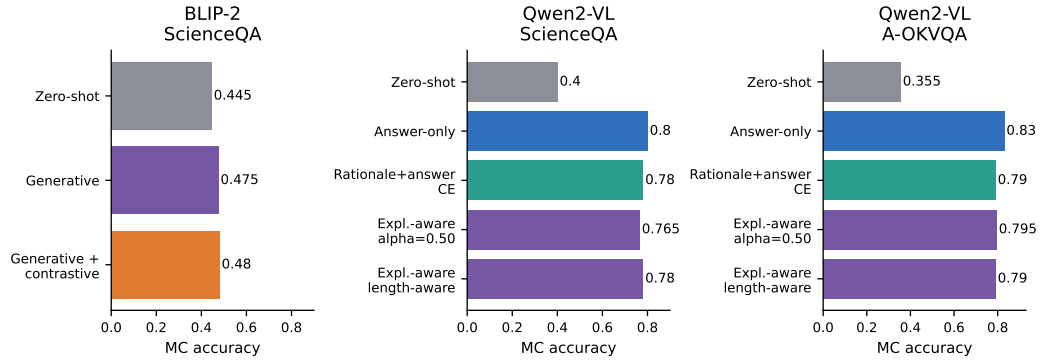


Figure 5: Controlled strategy comparison for BLIP-2 and Qwen2-VL. The figure shows zero-shot, answer-only, rationale-generative, explanation-aware, and contrastive variants where those controls are available.

412 *4.2. Effect of explanation-aware training*

413 Table 8 and Figure 6 trace the ScienceQA α -sweep, where α is the answer-
 414 span weight from Equation 1. Fine-tuning alone lifts accuracy from the
 415 0.400 zero-shot reference into the 0.690–0.780 band. The sweet spot is low:
 416 $\alpha = 0.10$, at 0.780. Push α toward the answer-heavy end and accuracy slides,
 417 to 0.690 at $\alpha = 0.75$ and 0.695 at $\alpha = 1.00$. The explanation metrics move
 418 in lockstep. ROUGE-L and BLEU also peak at $\alpha = 0.10$ (0.695 and 0.566)
 419 and decay as α climbs. At $\alpha = 1.00$ only the answer span is supervised
 420 inside a rationale-shaped target, so there are no rationale metrics to report.
 421 In accuracy, the $\alpha = 0.10$ run ties the rationale-generative and length-aware
 422 controls. One trap worth flagging: the $\alpha = 1.00$ row is not the true answer-
 423 only control, because the model still sees a rationale-shaped template even
 424 though the rationale span goes unsupervised. The genuine answer-only Sci-
 425 enceQA control, on its own prompt family, reaches 0.800. And all of this
 426 rests on one seed, 2,000 training and 200 evaluation examples, so these are
 427 observed effects, not final statistical proof.

Table 8: Explanation-aware fixed- α sweep on ScienceQA (Qwen2-VL-2B, QLoRA, 2,000 train / 200 eval, single seed). The coefficient α weights the answer span in Equation 1: $\alpha = 1$ is answer-span-only within the rationale target and $\alpha = 0$ is explanation-span-only. The zero-shot and rationale-generative rows are references. Random-choice accuracy is 0.354. The best fixed α is shown in bold.

Variant (α)	MC accuracy	ROUGE-L	BLEU
Zero-shot (frozen)	0.400	–	–
Rationale+answer CE	0.780	0.693	0.559
$\alpha = 0.00$ (explanation-only)	0.725	0.417	0.299
$\alpha = 0.10$	0.780	0.695	0.566
$\alpha = 0.25$	0.775	0.686	0.548
$\alpha = 0.50$	0.765	0.643	0.493
$\alpha = 0.75$	0.690	0.604	0.443
$\alpha = 1.00$ (answer-span only)	0.695	–	–

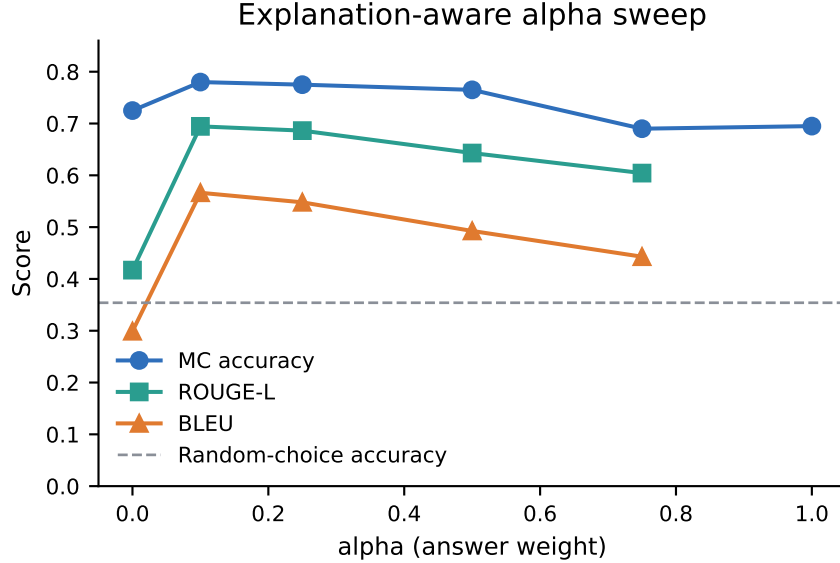


Figure 6: Fixed-alpha explanation-aware sweep on ScienceQA. Answer accuracy peaks at the low-answer-weight setting and declines toward both extremes inside the rationale-plus-answer target. Explanation quality (ROUGE-L, BLEU) peaks at the same point and falls as α increases. Series are distinguished by marker shape and line style; the dashed grey line marks random-choice accuracy. Single seed.

4.3. Domain-level behaviour

Table 9 and Figure 7 gather the selected domain runs from the final matrix. For ScienceQA the row uses the fixed $\alpha = 0.50$ explanation-aware run, so it lines up with the in-domain figure; the stronger $\alpha = 0.10$ and length-aware settings live in the sweep table above. The biggest jumps land on DocVQA, VQAv2, ChartQA, and A-OKVQA. A-OKVQA climbs from 0.355 to 0.795 accuracy against the reported zero-shot reference. The VQAv2 subset goes from 0.350 to 0.730 VQA accuracy. ChartQA reaches 0.655 relaxed accuracy, exact match 0.410. DocVQA is the headline, at 0.905 ANLS with exact match at 0.855. That number is strong for a capped internal run, but it deserves caution on three counts: the qualitative samples are short extractive answers, the setup is answer-only fallback rather than gold rationales, and the split audit cannot prove document-level disjointness past the native IDs it can see.

Table 9: Domain-level results for the completed Qwen2-VL-2B runs. Each domain uses its native headline metric: multiple-choice accuracy for ScienceQA and A-OKVQA, relaxed accuracy for ChartQA, ANLS for DocVQA, and VQA accuracy for the VQAv2 subset.

Domain	Dataset	Objective	Zero-shot ref.	Final	Secondary metric
Science	ScienceQA	Explanation-aware, $\alpha = 0.50$	0.400	0.765	ROUGE-L 0.643, BLEU 0.493
Commonsense	A-OKVQA	Explanation-aware, $\alpha = 0.50$	0.355	0.795	ROUGE-L 0.337, BLEU 0.087
Charts	ChartQA	Answer-only fallback	0.335	0.655	Exact match 0.410
Documents	DocVQA	Answer-only fallback	0.349	0.905	Exact match 0.855
Natural images	VQAv2 subset	Answer-only fallback	0.350	0.730	Exact match 0.350 $\rightarrow$ 0.730

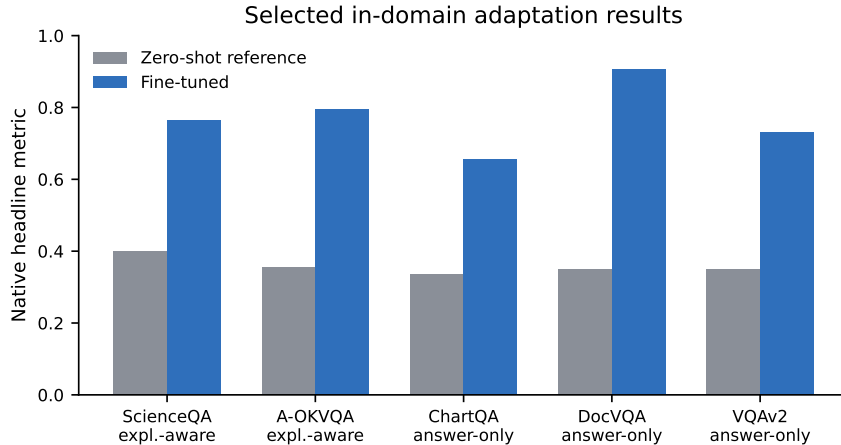


Figure 7: In-domain adaptation results for the selected completed Qwen2-VL-2B domain runs. The comparison mixes rationale-supervised ScienceQA/A-OKVQA rows with answer-only fallback rows for domains without gold rationales, so it should not be read as an explanation-aware effect across all domains.

4.4. Faithfulness and hallucination analysis

Table 10 and Figure 8 carry the evidence-masking drift check. Each number is a mean over 30 evaluation examples. The bigger and more positive it is, the more masking a region costs the gold-answer likelihood, which means the model was relying on what we hid. The drift piles up on the structured tasks: DocVQA at 0.375, ChartQA at 0.224, VQAv2 at 0.111. ScienceQA and A-OKVQA usually sit far lower, with one loud exception, the ScienceQA answer-only adapter at 0.0898. That low ScienceQA drift is not surprising; the questions come loaded with language priors and explicit answer choices, the visual evidence is diffuse, and a coarse mask can miss the single cue the model actually used. Explanation supervision does not move drift in any clean way here. At $\alpha = 0.10$ it is 0.0223, at $\alpha = 0.50$ it is 0.0197, and answer-span-only $\alpha = 1.00$ gives 0.0149, despite that run producing no supervised

455 rationale at all. On A-OKVQA the explanation-aware run drifts a touch
 456 more than the generative control, 0.0271 against 0.0245, but the intervals
 457 around both are wide. Read plainly, the masking result backs a domain-
 458 sensitivity story, not the sweeping claim that explanation-aware training by
 459 itself buys faithfulness. We report no manual hallucination rate, because we
 460 never ran a fixed annotation protocol for hallucination categories.

Table 10: Evidence-masking drift by completed run. Higher values indicate stronger sensitivity to masked visual evidence. The intervals are approximate 95% intervals from the generated masking artifact.

Run	Mean drift	Approx. 95% CI	Examples
ScienceQA answer-only	0.0898	0.0197–0.1830	30
ScienceQA rationale+answer CE	0.0217	-0.0100–0.0582	30
ScienceQA $\alpha = 0.10$	0.0223	-0.0080–0.0568	30
ScienceQA $\alpha = 0.50$	0.0197	-0.0119–0.0552	30
ScienceQA answer-span-only $\alpha = 1.00$	0.0149	-0.0274–0.0638	30
A-OKVQA rationale+answer CE	0.0245	-0.0236–0.0726	30
A-OKVQA explanation-aware	0.0271	-0.0075–0.0627	30
ChartQA answer-only fallback	0.2237	0.1419–0.3140	30
DocVQA answer-only fallback	0.3749	0.2770–0.4960	30
VQAv2 answer-only fallback	0.1109	0.0274–0.2408	30
Qwen2-VL-7B ScienceQA $\alpha = 0.50$	0.0114	-0.0059–0.0260	30

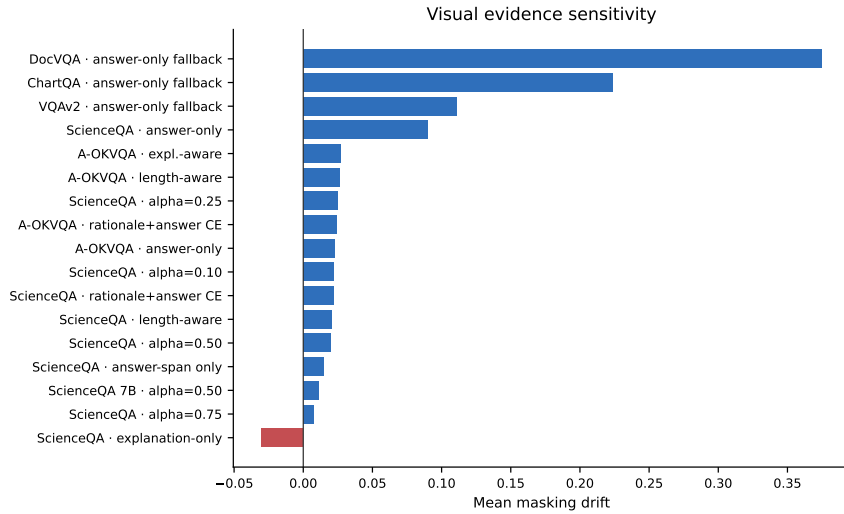


Figure 8: Mean evidence-masking drift across completed runs. Structured document and chart runs show much stronger visual sensitivity than the rationale-rich ScienceQA and A-OKVQA runs, but stronger drift does not automatically imply better answer accuracy.

ChartQA masking diagnostic: top-drift mask hurts the answer more than a random mask

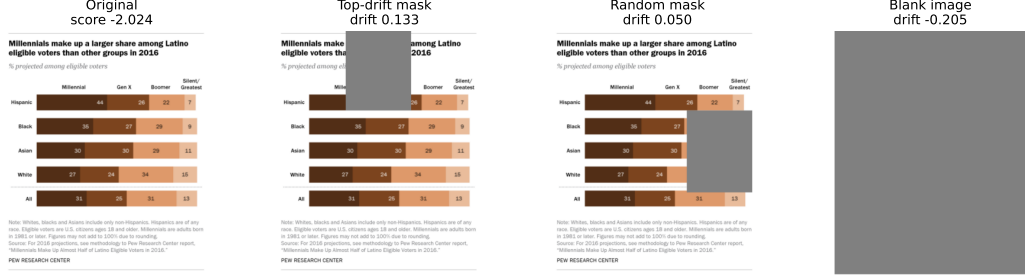


Figure 9: Example masking diagnostic for a ChartQA answer-only fallback run. The pipeline saves the original image, the top-drift mask, a random-region mask, and a blank-image control, together with the full-image and masked answer scores. This figure is used as an evidence-sensitivity example, not as proof that a generated rationale is faithful.

4.5. Efficiency and scale trade-offs

Table 11 and Figure 10 put Qwen2-VL-2B and Qwen2-VL-7B side by side on the same ScienceQA explanation-aware setting ($\alpha = 0.50$). Scaling up pays. The 7B run lifts accuracy from 0.765 to 0.885. ROUGE-L rises from 0.643 to 0.738, BLEU from 0.493 to 0.637. It is the single largest gain anywhere in the completed set. Both runs still fit under QLoRA. What the logs do not give us is a stable wall-clock or peak-memory figure, so this stands as a model-scale result, not a precise efficiency one.

Table 11: Qwen2-VL scale comparison on ScienceQA with the same explanation-aware objective ($\alpha = 0.50$).

Model	MC accuracy	ROUGE-L	BLEU
Qwen2-VL-2B	0.765	0.643	0.493
Qwen2-VL-7B	0.885	0.738	0.637

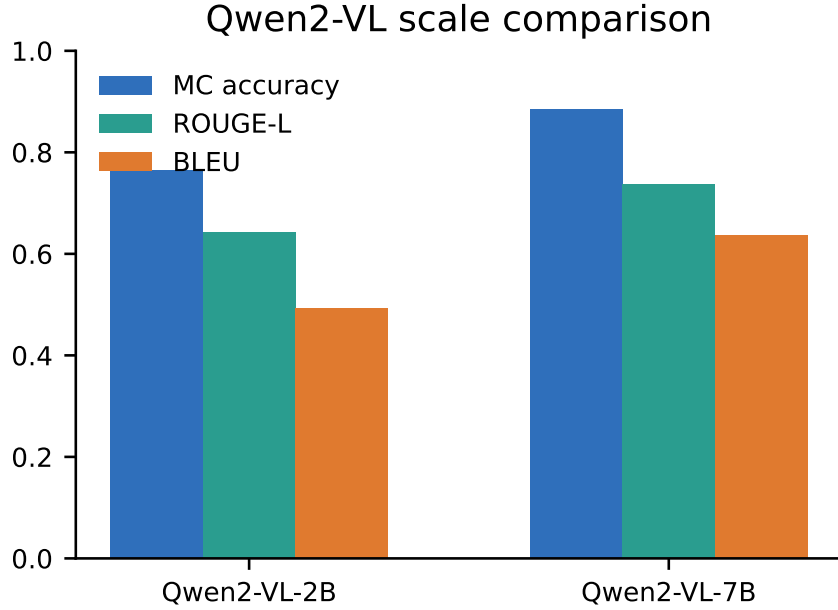


Figure 10: Scale comparison for the fixed ScienceQA explanation-aware setting. Moving from Qwen2-VL-2B to Qwen2-VL-7B improves both answer accuracy and rationale-overlap metrics.

4.6. Cross-domain transfer

Table 12 and Figure 11 lay out the train-domain against test-domain transfer matrix. Every column is scored in its target domain’s own metric, so the only safe comparisons run down a column, not across a row. Where the controls exist, the matrix keeps answer-only, rationale-generative, and explanation-aware sources separate. Treat it as an evaluation probe rather than the official in-domain table; a diagonal cell can drift a little from the matching completed run when the transfer evaluator routes through a different prompt family. The probe matters because answer-only training often travels better than rationale-generative training. The ScienceQA answer-only adapter is the clearest case, reaching 0.805 on A-OKVQA, 0.645 on ChartQA, 0.703 on DocVQA, and 0.630 on VQAv2. Its rationale-generative and explanation-aware siblings clear the random baselines but fall off on ChartQA, DocVQA, and VQAv2. A-OKVQA sources split the honours: rationale-generative training wins DocVQA transfer in that group at 0.764, while answer-only wins VQAv2 at 0.700. DocVQA is strongest at home (0.910), and the document-trained adapter carries over to ChartQA too

486 (0.685). The pattern, in short: transfer depends on the target domain and
 487 the source objective together, not on domain alone.

Table 12: Cross-domain transfer matrix. Rows indicate the training domain and columns indicate the evaluation domain. Each column uses the target domain’s headline metric, so values are best compared within a column.

Trained on	ScienceQA	A-OKVQA	ChartQA	DocVQA	VQAv2
ScienceQA answer-only	0.800	0.805	0.645	0.703	0.630
ScienceQA rationale CE	0.780	0.760	0.450	0.559	0.300
ScienceQA explanation-aware	0.740	0.765	0.460	0.491	0.410
A-OKVQA answer-only	0.690	0.825	0.635	0.677	0.700
A-OKVQA rationale CE	0.665	0.790	0.645	0.764	0.600
A-OKVQA explanation-aware	0.680	0.795	0.615	0.763	0.640
ChartQA answer-only	0.420	0.415	0.660	0.831	0.680
DocVQA answer-only	0.450	0.435	0.685	0.910	0.660
VQAv2 answer-only	0.440	0.575	0.660	0.716	0.730

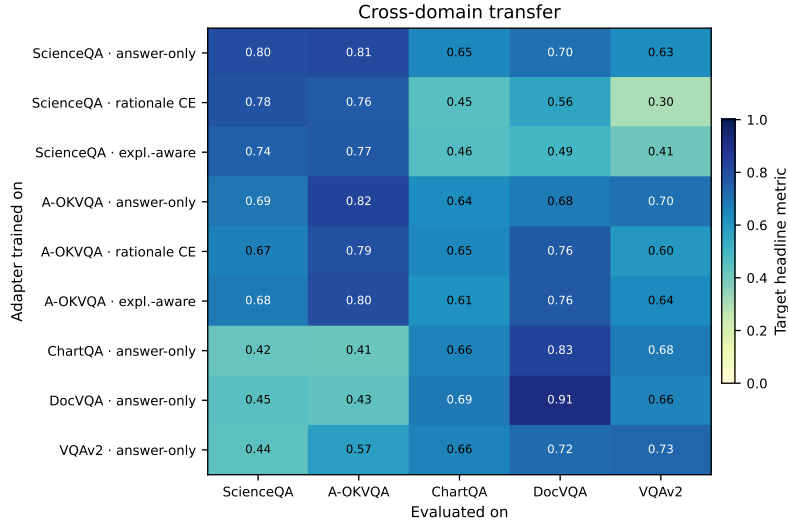


Figure 11: Cross-domain transfer heatmap. The strongest non-diagonal transfer is not always from the visually closest domain. Answer-only ScienceQA transfers strongly to A-OKVQA, while answer-only document and chart adapters transfer strongly to several open-ended targets.

488 5. Discussion

489 The results back the central idea, but only with caveats attached. They
490 also slot neatly into the arc the literature review traced: VLM alignment has
491 drifted from image–text representation matching toward generative instruc-
492 tion following, harder multimodal reasoning tests, and separate audits for
493 hallucination, critique, and evidence use. Objective choice clearly matters;
494 explanation supervision being automatically better than plain answer-only
495 fine-tuning does not follow. It can help, yes, but only when it is balanced with
496 care and held against a true answer-only control. On ScienceQA the best
497 fixed explanation-aware run is the low-answer-weight one, and the length-
498 aware control merely matches the rationale-generative control. The answer-
499 span-only setting is plainly worse on both accuracy and rationale quality.
500 A-OKVQA shows a small observed edge for explanation-aware training over
501 the generative control, but the capped-run uncertainty is wide enough that
502 we will not call it a firm win. And the fixed $\alpha = 0.50$ ScienceQA run is
503 actually worse than both the rationale-generative and answer-only controls.
504 Adding explanation tokens, by itself, does not make the model better. The
505 loss balance is what does the work.

506 The BLIP-2 contrastive result earns attention precisely because it is so
507 weakly positive on accuracy. We built the contrastive-enhanced arm specifi-
508 cally so the comparison could be run head-on. In the completed ScienceQA
509 run it sits just 0.005 above the BLIP-2 generative control, well inside the
510 approximate confidence intervals. Retrieval barely separates from the frozen
511 and random controls either. The read is that the auxiliary image–answer In-
512 foNCE term, at least in this small Q-Former-side setup, cannot forge a clearly
513 stronger image–answer representation. For this experiment set, generative
514 and explanation-aware supervision rest on firmer ground than contrastive
515 enhancement.

516 The domain results are the clearest argument against trusting a single
517 benchmark. ChartQA, DocVQA, VQAv2, A-OKVQA, and ScienceQA every
518 one improves in the selected runs. Yet the reading stays mixed, because the
519 domains are not all testing the same kind of supervision. ScienceQA and A-
520 OKVQA carry rationale-bearing targets; ChartQA, DocVQA, and VQAv2
521 fall back to answer-only training, because no gold rationales are there to
522 use. The strong DocVQA number, then, is capped extractive document
523 answering, nothing more, not evidence of faithful document rationales and
524 not a public-test DocVQA claim. Any stronger document-reasoning claim

would first need document-specific rationale supervision, or a more causal OCR and layout faithfulness test.

The faithfulness numbers come with a warning attached. High masking drift proves the model is sensitive to visual change; it does not prove the explanation is faithful. DocVQA and ChartQA drift hard because they live on local visual evidence, and in this run both also improve under plain answer-only fallback. ScienceQA and A-OKVQA post better answer and rationale scores, yet their drift stays small and refuses to climb consistently with explanation-aware training. Better rationale text, in other words, is not the same animal as more faithful visual grounding. So the masking diagnostic earns a place beside BLEU and ROUGE-L, never a place instead of them.

Scale hands us the cleanest positive result of all. Holding the ScienceQA explanation-aware objective fixed, Qwen2-VL-7B beats the 2B model on both accuracy and rationale metrics. That does not retire the 2B experiments, since the small model is exactly what made the wide sweep affordable in the first place. What it does say is that some of the 2B weakness is probably about capacity, not objective.

Transfer is mixed in the same way. Answer-only ScienceQA jumps cleanly to A-OKVQA, while the rationale-generative and explanation-aware ScienceQA adapters lose ground on the open-ended targets. A-OKVQA repeats the shape: rationale-bearing objectives help on some targets, but answer-only is the stronger bet for VQAv2. The answer-only ChartQA and DocVQA adapters trade well with each other under the target metrics, a hint that structured, text-heavy images share something useful when it comes to adaptation. The conclusion is the same one again: transfer rides on the target domain and the source objective together.

5.1. Limitations

Compute is the first limitation, and the obvious one. A single-GPU budget keeps the study realistic, but it also caps the seeds, the model sizes, and the full-dataset runs we could attempt. Rationale availability is the second. ScienceQA and A-OKVQA ship gold explanations; DocVQA, ChartQA, and VQAv2 do not, so they cannot carry the same rationale supervision, and any teacher-generated rationale experiment on them must be reported apart from gold-rationale training. The metrics are the third. BLEU and ROUGE-L can wave off a perfectly valid explanation just because the wording differs, and while the masking metric gets closer to actual evidence use, it hinges on

562 which regions are masked and turns noisy on dense documents and charts.
 563 The protocol is the fourth: capped train and evaluation subsets, one seed
 564 per full configuration. That is what makes the work feasible and auditable,
 565 but it also means small gaps are not statistically settled. And finally, the
 566 execution logs are good enough to validate the outputs, yet not good enough
 567 for precise wall-clock or peak-memory benchmarking. Taken together, these
 568 are the lines the paper will not cross in what it claims.

569 5.2. Future directions

570 The first thing future work should do is rerun the strongest configura-
 571 tions with more seeds and larger evaluation splits. After that, swap in newer
 572 open multimodal backbones while keeping the objective-isolation protocol
 573 untouched. A third push is better rationale annotation for chart and doc-
 574 ument datasets, the very places where faithful explanations would pay off
 575 most. The faithfulness test itself should grow more causal, editing document
 576 text, chart marks, or image regions in controlled ways rather than mask-
 577 ing blindly. There is also a clean comparison waiting between supervised
 578 explanation-aware training and preference-based or reinforcement-learning
 579 methods that reward grounding directly. And before any of this ships, de-
 580 ployment work should track latency, memory, and calibration alongside ac-
 581 curacy and faithfulness.

582 6. Conclusion

583 This paper asked what the alignment objective alone does to vision-
 584 language reasoning under a fixed, resource-constrained fine-tuning bud-
 585 get. The comparison spanned zero-shot baselines, BLIP-2 generative and
 586 contrastive-enhanced alignment, and Qwen2-VL in answer-only, rationale-
 587 generative, and explanation-aware forms, across document, chart, science,
 588 commonsense, and natural-image benchmarks. Throughout, we mea-
 589 sured answer accuracy, rationale quality, and evidence sensitivity together
 590 rather than one at a time. What holds up best in the completed runs is
 591 Qwen2-VL task adaptation: rationale supervision applied with care where
 592 gold rationales exist, answer-only fallback where they do not. BLIP-2
 593 contrastive-enhanced training is here as a route we genuinely tested, but
 594 its gain over BLIP-2 generative training is small and its retrieval diagnostic
 595 is weak. All five datasets, ScienceQA, A-OKVQA, ChartQA, DocVQA,
 596 and VQAv2, improve under the selected runs. The faithfulness analysis

lands on a distinction worth keeping: better explanation text and stronger visual evidence use are related, not identical. Explanation-aware training lifts rationale text in some settings, yet it does not reliably raise masking-based evidence dependence. Scale is the cleanest win, with Qwen2-VL-7B strengthening the ScienceQA explanation-aware result, while the transfer matrix shows alignment staying stubbornly target-domain dependent. The honest summary is a careful one: supervised rationales can help resource-constrained VLMs, but only when answer-only controls, domain shift, and the evidence test are weighed together.

Declarations

Author responsibility. The authors wrote, reviewed, and edited the manuscript and take full responsibility for the final content, results, and claims.

Use of generative AI. Generative AI tools were used only for language editing, code assistance, and consistency checks. All scientific decisions, experiment selection, result interpretation, and final manuscript approval remain the responsibility of the authors.

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Data, code, and artifacts. The experiments use public datasets cited in the paper. The code, configuration files, generated tables, figures, execution logs, and claim-to-evidence artifacts are maintained with the project repository and can be made available with the submitted supplementary material.

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